Human - AI detection

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Why Detect AI-Generated Text?

Combating Disinformation (Society)

Problem: AI can generate massive amounts of **fake news** and malicious comments at low cost.

Goal: Prevent social manipulation and protect democratic decision-making by filtering AI-bots.

Related Work

1. **RoBERTa** : updates model parameters with **pre-training**

1. **Fast-DetectGPT** : keeps parameters fixed and relies solely on **prompts(zero-shot)**. faster version of DetectGPT

Liu et al. (2019). RoBERTa: A Robustly Optimized BERT Pretraining Approach.
Bao et al., "Fast-DetectGPT: Efficient Zero-Shot Detection of Machine-Generated Text via Conditional Probability Curvature," *ICLR*, 2024.

Core Hypothesis of DetectGPT

Probability Curvature

- Model-generated text tends to occupy negative curvature regions of the model's log-probability function.
- Essentially, the generated text is a **local maximum** in terms of probability.

Limitation

- Generates only surface-level variations due to heavy reliance on preserved local context.

Mitchell et al., "DetectGPT: Zero-Shot Machine-Generated Text Detection using Probability Curvature," *ICML*, 2023.

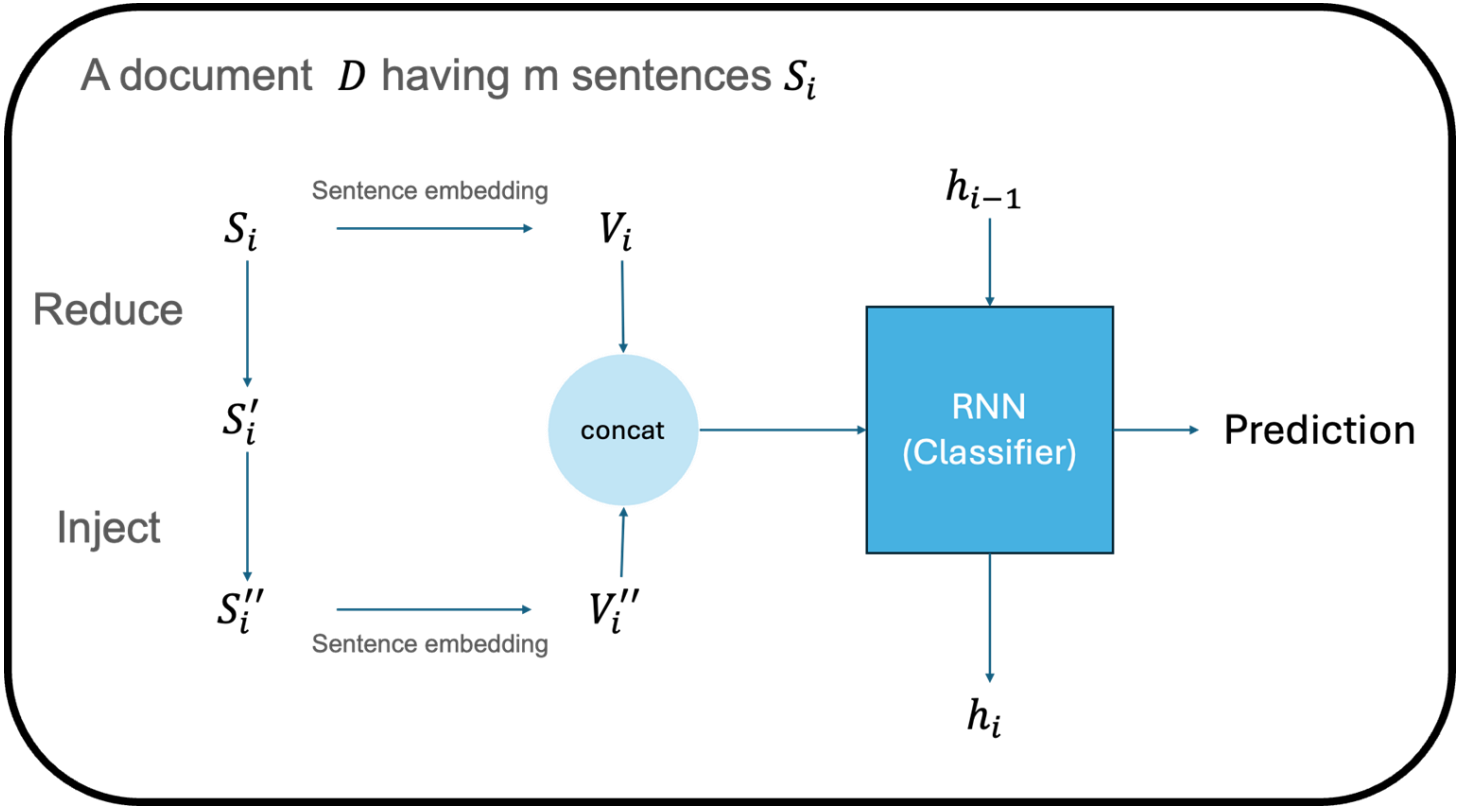
Our Model

1. Do zero-shot to extract argument and discard predicate from the sentence. (prevent information leakage)

1. Do zero-shot to inject predicates into argument. (maximize log probability)

1. Check probability difference between original document, and injected document to distinguish human and AI.

Method


A document D having m sentences S_i
Sentence embedding
S_i
V_i
h_{i-1}
Reduce
S_i'
concat
RNN
(Classifier)
Prediction
Inject
S_i''
V_i''
Sentence embedding
h_i

Sentence Embedding Construction

1. Token Embedding

- Convert input tokens into dense vector representations.

2. Positional Embedding

- Add positional encodings to capture the order of tokens.

3. Mean Pooling (Sentence Representation)

- Compute the average of all token vectors to derive the final sentence embedding.

Experiment Result (llm-generated-texts)

	Accuracy	Precision	Recall	F1-Score
our model	**97.2**	**98.3**	**96.1**	**97.2**
RoBERTa	68.3	87.4	42.9	57.5
Fast-DetectGPT	57.8	62.1	39.8	48.5

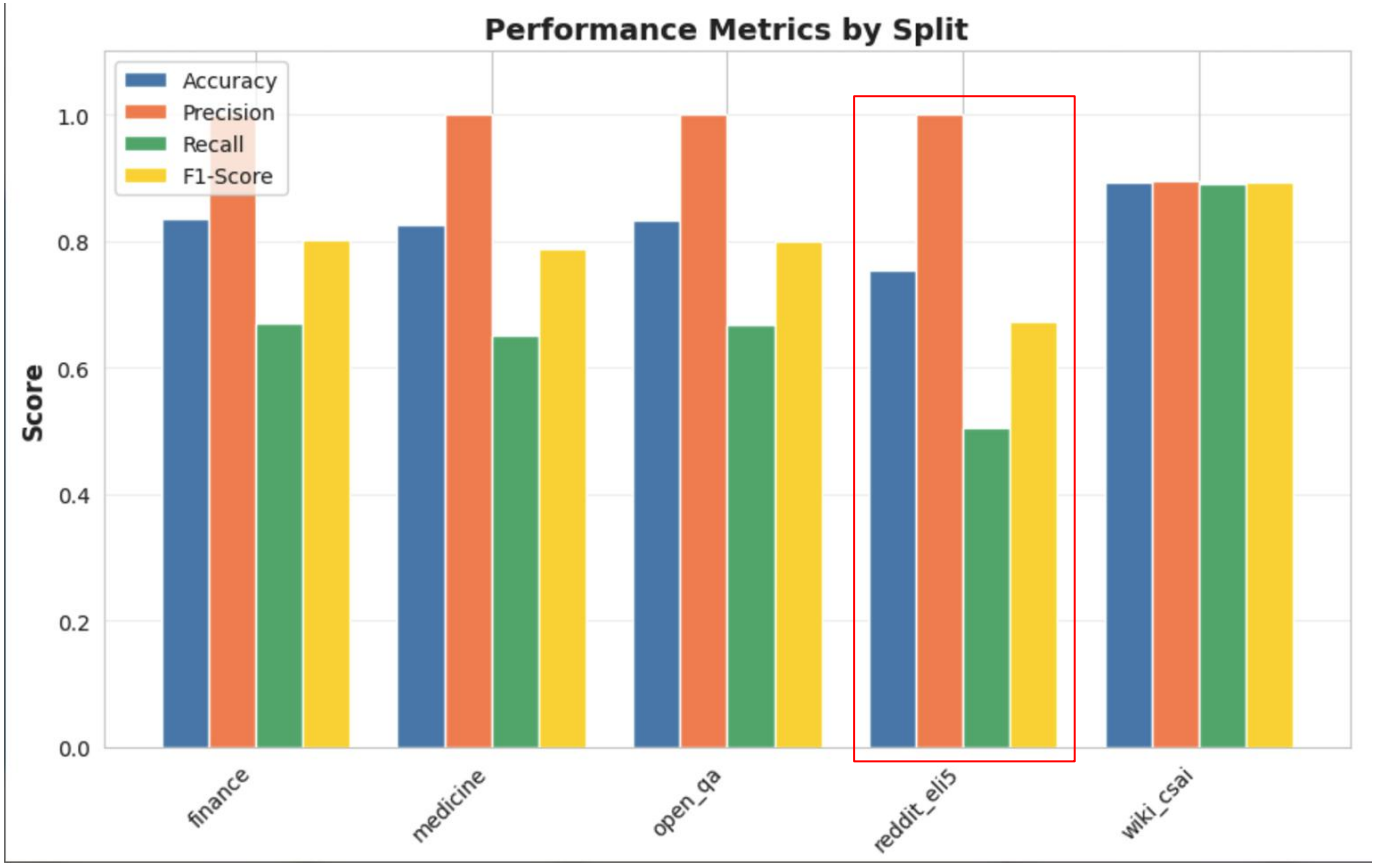
Experiment Result (HC3)

	finance	medicine	open_qa	reddit_eli5	wiki_csai
our model	83.5	82.5	**83.3**	75.3	89.3
RoBERTa	**92.5**	**94.0**	78.5	**97.0**	**95.5**
Fast-DetectGPT	74.5	76.5	80.0	82.0	79.0

Accuracy

Performance of ours on HC3

very low recall



Limitation

Cost of reducing - injecting is expensive

Future Work

Selection of the optimal classifier component by evaluating backbone generalization on OOD datasets

Thank You